



R Packages

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What do we want to be able to do?

Data Science!

- Read in raw data and manipulate it
- Combine data sources
- Summarize data to glean insights
- Apply common analysis methods
- Communicate Effectively

Where Do Our Objects & Functions Come From?

When you open R a few `packages` are loaded

- R package:
 - Collection of functions/objects/datasets/etc.

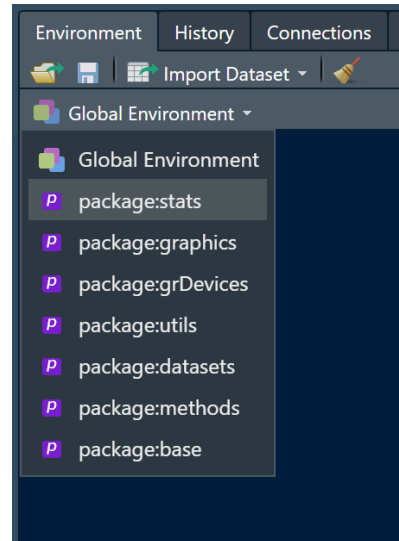
Where Do Our Objects & Functions Come From?

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- R package:
 - Collection of functions/objects/datasets/etc.
- Packages exist to do almost anything
 - [List of CRAN](#) approved packages
 - Plenty of other packages on places like GitHub!

Where Do Our Objects & Functions Come From?

- When you open R a few `packages` are loaded

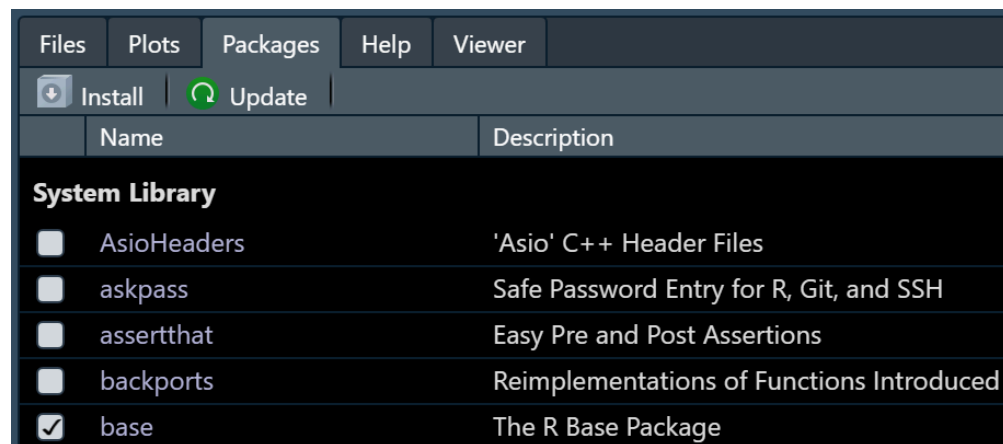


- `base` package has `c()`, `data.frame()`, `list()`, ...

Installing an R Package

- First time using a package
 - Must **install package**
 - Can use code, menus, or Packages tab

```
install.packages("dplyr")
```



Accessing a Package in Your R Session

- Only install once!
- **Each session:** read in package using `library()` or `require()`

```
library("dplyr")
```

Accessing a Package in Your R Session

- Difference - if no package
 - `library()` throws an error
 - `require()` returns FALSE

```
library("notAPackage")
```

```
## Error in `library()`:  
## ! there is no package called 'notAPackage'
```

```
require("notAPackage")
```

```
## Warning in library(package, lib.loc = lib.loc, character.only = TRUE,  
## logical.return = TRUE, : there is no package called 'notAPackage'
```


Setting Packages to Automatically Load

To load packages by default, access your `.Rprofile` file

- Add in the code

```
options(defaultPackages=c(getOption("defaultPackages"),  
                           "mypackage1",  
                           "mypackage2"))
```

See [here](#) for info on finding your `.Rprofile` file

Accessing a Package in Your R Session

- See everything from a package:

```
ls("package:dplyr")
```

```
## [1] "%>%"           "across"          "add_count"
## [4] "add_count_"    "add_row"         "add_rownames"
## [7] "add_tally"     "add_tally_"      "all_equal"
## [10] "all_of"        "all_vars"        "anti_join"
## [13] "any_of"        "any_vars"        "arrange"
## [16] "arrange_"      "arrange_all"     "arrange_at"
## [19] "arrange_if"    "as.tbl"          "as_data_frame"
## [22] "as_label"      "as_tibble"       "auto_copy"
## [25] "band_instruments" "band_instruments2" "band_members"
## [28] "bench_tbls"    "between"         "bind_cols"
## [31] "bind_rows"     "c_across"        "case_match"
## [34] "case_when"     "changes"         "check_dbplyr"
## [37] "coalesce"      "collapse"        "collect"
## [40] "combine"       "common_by"       "compare_tbls"
## [43] "compare_tbls2" "compute"         "consecutive_id"
## [46] "contains"      "copy_to"         "count"
## [49] "count_"       "cross_join"
```

Calling From a Library

- Call functions without loading full library with `::`
- If not specified, most recently loaded package takes precedent

```
#stats::filter(...) calls time-series function from stats package  
dplyr::filter(iris, Species == "virginica")
```

```
##      Sepal.Length Sepal.Width Petal.Length Petal.Width  Species  
## 1           6.3         3.3         6.0         2.5 virginica  
## 2           5.8         2.7         5.1         1.9 virginica  
## 3           7.1         3.0         5.9         2.1 virginica  
## 4           6.3         2.9         5.6         1.8 virginica  
## 5           6.5         3.0         5.8         2.2 virginica  
## 6           7.6         3.0         6.6         2.1 virginica  
## 7           4.9         2.5         4.5         1.7 virginica  
## 8           7.3         2.9         6.3         1.8 virginica  
## 9           6.7         2.5         5.8         1.8 virginica  
## 10          7.2         3.6         6.1         2.5 virginica  
## 11          6.5         3.2         5.1         2.0 virginica  
## 12          6.4         2.7         5.3         1.9 virginica  
## 13          6.8         3.0         5.5         2.1 virginica  
## 14          5.7         2.5         5.0         2.0 virginica  
## 15          5.8         2.8         5.1         2.4 virginica  
## 16          6.4         2.8         5.3         2.3 virginica  
## 17          6.5         2.9         5.5         1.8 virginica  
## 18          6.7         2.2         6.7         2.2 virginica  
## 19          6.9         2.3         6.9         2.3 virginica  
## 20          6.0         2.2         5.0         1.5 virginica
```

Quick R Example

Let's jump into R and see how to download a package that will allow us to create a `.pdf` file from our `.qmd`!

R Packages Recap

- Install packages first (download it)
 - Can do more than one at a time
- Load package with `require()` or `library()`
- Call a function without loading using `::`